



Test Report - TIA-Demo Prototype-001

Test Information

Project: TIA-Demo DIP Configuration
Operator: Tobias-NUC
Date: 20.09.2026
Time: 17:00:00
Duration: 30,14s

Device Under Test

Serial Number: Prototype-001
Device Name: TIA-Demo
Revision: B
Variant: I2C DAC + MCU
Part Number: -

Additional Notes:

DIP switches determine which tests are executed

PASS

Step	Name	Lower Limit	Measured	Upper Limit	Result
2	VIN 12V?	10,8V	11,992V	13,2V	✓
3	U1 VOUT 3V3?	2,97V	3,231V	3,63V	✓
4	D2 Forward Voltage	1,71V	2,017V	2,09V	✓
5	R1 Voltage Drop	1,17V	1,289V	1,43V	✓
6	D2 Forward Current	1,17mA	1,289mA	1,43mA	✓
10	Segment A ON?	2,97V	3,209V	3,63V	✓
12	Segment B ON?	2,97V	3,184V	3,63V	✓
14	Segment C ON?	2,97V	3,196V	3,63V	✓
16	Segment D ON?	2,97V	3,258V	3,63V	✓
18	Segment E ON?	2,97V	3,144V	3,63V	✓
20	Segment F ON?	2,97V	3,182V	3,63V	✓
22	Segment G ON?	2,97V	3,179V	3,63V	✓
23	### Read DIP Switches ###	0	0	255	✓
24	### DIP Switch 1 ON? ###	0	0	0	✓
27	D3 Red Forward Voltage	1,53V	1,764V	1,87V	✓
28	R22 Voltage Drop	1,35V	1,509V	1,65V	✓
29	D3 Red Forward Current	1,143mA	1,258mA	1,397mA	✓
31	D3 Green Forward Voltage	2,07V	2,253V	2,53V	✓
32	R23 Voltage Drop	900mV	986,085mV	1,1V	✓
33	D3 Green Forward Current	747uA	821,738uA	913uA	✓



Test Report - TIA-Demo Prototype-001

Step	Name	Lower Limit	Measured	Upper Limit	Result
35	D3 Blue Forward Voltage	2,25V	2,514V	2,75V	✓
36	R24 Voltage Drop	675mV	773,723mV	825mV	✓
37	D3 Green Forward Current	576uA	644,769uA	704uA	✓
38	### DIP Switch 2 ON? ###	0	0	0	✓
41	Button 1 High?	2,97V	3,223V	3,63V	✓
42	Button 1 Low?	-200mV	1,833mV	200mV	✓
43	Button 2 High?	2,97V	3,237V	3,63V	✓
44	Button 1 Low?	-200mV	1,001mV	200mV	✓
45	### DIP Switch 3 ON? ###	0	0	0	✓
48	Divider with 680R	189mV	209,17mV	231mV	✓
49	Divider with 1k2	315mV	354,039mV	385mV	✓
50	Divider with 2k2	531mV	593,562mV	649mV	✓
51	Divider with 3k9	828mV	923,9mV	1,012V	✓
52	Divider with 6k8	1,17V	1,332V	1,43V	✓
53	### DIP Switch 4 ON? ###	0	0	0	✓
56	POT1 Adjusted to 1,5V	1,45V	1,513V	1,55V	✓
57	### DIP Switch 5 ON? ###	0	0	0	✓
60	Temperature Sensor VOUT	693mV	703,025mV	847mV	✓
61	Temperature	18°C	20,303°C	32°C	✓
62	### DIP Switch 6 ON? ###	0	0	0	✓
64	Set I2C DAC to 0.5V	1	1	1	✓
65	Measure DAC Output 0.5V?	450mV	499,02mV	550mV	✓
66	Set I2C DAC to 1.0V	1	1	1	✓
67	Measure DAC Output 1.0V?	900mV	973,705mV	1,1V	✓
68	Set I2C DAC to 1.5V	1	1	1	✓
69	Measure DAC Output 1.5V?	1,35V	1,477V	1,65V	✓
70	Set I2C DAC to 2.0V	1	1	1	✓
71	Measure DAC Output 2.0V?	1,8V	1,978V	2,2V	✓
72	Set I2C DAC to 2.5V	1	1	1	✓
73	Measure DAC Output 2.5V?	2,25V	2,472V	2,75V	✓
74	Set I2C DAC to 3.0V	1	1	1	✓
75	Measure DAC Output 3.0V?	2,7V	2,961V	3,3V	✓
76	### DIP Switch 7 ON? ###	0	0	0	✓



Test Report - TIA-Demo Prototype-001

Step	Name	Lower Limit	Measured	Upper Limit	Result
78	Program MCU	0	0	0	✓
79	Get FW-Version	1,2	1,2	1,2	✓
80	### DIP Switch 8 ON? ###	0	0	0	✓
83	Read POT2 Value from MCU	1450mV	1517mV	1550mV	✓
84	Configure MCU DAC to 0.5V	1	1	1	✓
85	Read DAC Output from MCU ADC	450mV	500mV	550mV	✓
86	Measure DAC Output Voltage	450mV	500,912mV	550mV	✓
87	Configure MCU DAC to 1.0V	1	1	1	✓
88	Read DAC Output from MCU ADC	950mV	995mV	1050mV	✓
89	Measure DAC Output Voltage	950mV	978,109mV	1,05V	✓
90	Configure MCU DAC to 1.5V	1	1	1	✓
91	Read DAC Output from MCU ADC	1450mV	1492mV	1550mV	✓
92	Measure DAC Output Voltage	1,45V	1,477V	1,55V	✓
93	Configure MCU DAC to 2.0V	1	1	1	✓
94	Read DAC Output from MCU ADC	1950mV	1989mV	2050mV	✓
95	Measure DAC Output Voltage	1,95V	1,969V	2,05V	✓
96	Configure MCU DAC to 2.5V	1	1	1	✓
97	Read DAC Output from MCU ADC	2450mV	2485mV	2550mV	✓
98	Measure DAC Output Voltage	2,45V	2,464V	2,55V	✓
99	Configure MCU DAC to 3.0V	1	1	1	✓
100	Read DAC Output from MCU ADC	2950mV	2979mV	3050mV	✓
101	Measure DAC Output Voltage	2,95V	2,964V	3,05V	✓

Used Equipment

Test Hardware:

Device Name: compact Test Interface Adapter
Serial Number: 2026001
Firmware Version: 0.5.3

Instruments:

Rigol DMM: Rigol Technologies,DM858,DM8A274201911,00.01.00.00.22